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Questions to Ask Yourself before you Upgrade

How old is my PC?

If your PC is about a year or so old then you have a fair

chance of upgrading your processor without going through major changes.

Will the upgrade demand a new motherboard?

If you are using a Slot interface board then you may have trouble upgrading since socket adapters for the newer processors such as the Athlon XP and the Pentium III (Tualatin) are not available. Even if your motherboard sports the slower Pentium 4s (1.4 to 1.8 GHz) and uses the Socket 423 format, you will not be able to upgrade to the latest Pentium 4s since they use the newer Socket 478 format. If you have an older board and are using an older Athlon (200 MHz FSB) then you will not be able to upgrade to the newer Athlons that sport a 266 MHz FSB.

Which memory standard do I need?

You will need faster memory standards such as RDRAM for the Pentium 4 or DDR SDRAM for the Athlon XP. Most new applications and games consume a lot of RAM and 256 MB should be considered the bare minimum.

Does the upgrade involve changing the cabinet and power supply?

Pentium 4 requires an entirely new type of power supply standard (ATX 2.03), so you will need a Pentium 4 certified cabinet and power supply. An ATX 2.03 compliant power supply can be easily identified by the addition of a 2x2 pin connector and an optional 1x6 pin auxiliary connector, which looks like the old-style AT connector. Both are essential for running the Pentium 4. For the Athlon series you need a decent 300+ Watt power supply and a standard cabinet.

How much extra performance will my upgraded processor deliver?

Monitor your system's memory and CPU usage when running the applications you use and prefer, on your current setup. If you find that you consistently experience sluggish behaviour and your memory and CPU usage is at all times over 75 per cent, then you should consider an upgrade.

Am I ready for upgrade?

If you use graphics-rich applications and have a system that is over a year old, you will find significant improvement in performance if you upgrade to the current generation of processors. On the other hand, if you have a 500 MHz or higher processor and use your PC primarily for surfing the Internet and for office applications, then just adding more RAM will give you enough of a performance boost.

and, of course, gaming—will really appreciate the incredible power that the Elite category of processors can deliver.

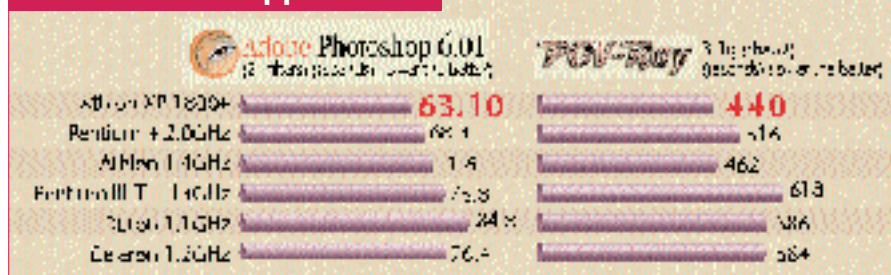
With the heated competition between AMD and Intel, processors have become much more powerful and affordable across all categories. Isn't it amazing that today, even what we call a value processor, runs at over 1 GHz and costs Rs 6,000? Competition has driven the prices down. The processors are much cheaper today when compared to their predecessors.

THE WINNERS

Athlon XP 1800+ is the undisputed champion and it beat the Pentium 4, 2 GHz in almost all the tests by a fair margin. Despite all the hype behind Pentium 4, 2 GHz, it fails to live up to the performance of Athlon XP 1800+ in key benchmarks like Photoshop, Business Winstone 2001 and 3DMark2001.

Choosing a winner for the Best Value award was much more difficult. While the Duron has consistently outperformed the Celeron by a mile in the past, the latest Celeron (Tualatin) 1.2 GHz processor, which made it into our labs just in time for this comparison test, matched up well against the 1.1 GHz Duron in some tests, and even beat it in the Business Winstone 2001 by a large margin. Despite this, we voted the Duron 1.1 GHz as our Best Value processor because it's Rs 1,000

Professional Applications



The less time taken the better

less than the Celeron, performs really good, and drives a fair bargain.

Athlon XP (Palamino)

The Athlon XP is AMD's latest warhorse in the fight for processor supremacy. Based on the Palamino core, it can push core clock speed up to 1.6 GHz (Athlon XP 1900+). The Athlon XP builds upon the strengths of the AMD Athlon series and is a worthy competitor of the Pentium 4.



The Athlon XP 1800+ tested in this comparison test simply blew the Pentium 4, 2 GHz to bits by completing the POV-Ray FPU intensive ray-tracing test in just 7 minutes 2 seconds.

The Pentium 4, 2 GHz took over

1 minute 34 seconds extra to complete this test. All of this goes to show that the real meat lies in the performance and not in the 'speed'.

In the video encoding test, the Athlon XP 1800+ took just 99 seconds to complete the test as compared to 111 seconds by the Pentium 4, 2 GHz.

This translates into a performance difference of over 10 per cent with a MHz deficiency of over 400 MHz!

Rating: A+

Verdict: If you want the fastest in terms of pure performance, the Athlon XP 1800+ will deliver.

Duron (Morgan)

The Duron series is essentially a low-cost variant of the Athlon range of chips. The Morgan core is based on the Palamino, which appears in the Athlon XP. The Duron 1.1 GHz (Morgan) beat the Pentium III 1.13 GHz (Tualatin) in the ray-tracing test: it scored 9 minutes